



Tecnologías Altur S.A.P.I. de C.V.
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HackMTY26 Challenge Track

Defend the Bank Against Voice Deepfakes

About Altur

Altur builds AI voice agents that handle millions of phone calls for Latin America's largest financial institutions. Our platform powers collections, customer service, and outbound campaigns for banks across Mexico, Colombia, Perú, Chile and Brazil, real conversations, in Spanish, over regular phone lines.

The problem today

Voice used to be proof of identity. Not anymore. With a few seconds of public audio, anyone can clone a voice well enough to fool both people and machines.

Contact centers are the front door of every bank, and they are now a target from both directions: fraudsters impersonate customers to take over accounts, and impersonate banks to social-engineer customers into handing over credentials. The people most exposed are the ones who bank by phone because they can't or don't use apps.

Today, most contact centers, human-staffed or AI-powered, have no solid way to tell whether the voice on the line belongs to a real person.

The big challenge

How can a bank's phone line know, in real time, that it's talking to a real human?

Build a system that detects, resists, or defends against synthetic voices in phone conversations.

Possible approaches

1. **Acoustic detection.** the voice itself. Train or fine-tune a classifier on the audio: spectral artifacts, prosody, breathing, model fingerprints.
2. **Conversational behavior.** how the caller handles the flow of a real conversation. Every call in the dataset includes moments where our agent interrupts, falls silent, or talks over the caller. Humans recover from these instantly and messily. Machines recover consistently, and consistency is a signal.

3. **Semantic.** what the caller actually says. Our agent asks callers to repeat information back, and sometimes asks about things that do not exist. A person says "I don't have that." A language model tends to invent an answer.

These are directions, not requirements. The strongest submissions will likely combine more than one, and ideas we haven't thought of score high on originality.

Resources provided

The recording dataset. Several hundred recorded phone conversations between a caller and an AI voice agent, in Spanish, released to teams at the start of the event. In some calls the caller is a real human; in others, a synthetic voice. Specifically:

- ❖ Stereo, telephony-grade 8kHz. Channel 0 is the incoming caller, the one you need to classify. Channel 1 is the agent.
- ❖ Diverse by design: a wide range of speakers, devices, accents, and call conditions.

Altur engineers on-site. The people who run these systems in production for real banks. Come talk to us.

What do build

A working prototype that takes a recorded phone conversation and decides whether the incoming caller is a real human or a synthetic voice.

To be scored in the automated round, your system must expose one HTTP endpoint:

POST /detect: receives a stereo WAV clip (8kHz, base64-encoded; channel 0 = caller, channel 1 = agent) and returns a verdict on the caller:

```
{
  "is_synthetic": true,
  "confidence": 0.87
}
```

is_synthetic is required. **confidence** is optional but recommended, we use it to break ties and to reward well-calibrated systems.

Everything else is yours to choose: model, language, framework, hosting. The endpoint only needs to be reachable during the scoring window. Include a short README explaining your approach.

Two things worth knowing. First, you get both sides of the conversation for a reason, the caller's audio is not the only signal available to you, and the agent's turns tell you what the caller was reacting to. Second, depth beats breadth. One signal, done well, will outscore three that half-work.

Evaluation format

Judges will visit each team for 15 minutes. During that time we run our benchmark against your endpoint using a hidden test set, and you walk us through your solution. Your endpoint must be reachable throughout judging.

Why it matters

The phone is still how a large part of Latin America banks, and for many customers it is the only channel they have. As synthetic voices become cheap and convincing, that channel either becomes verifiable or becomes untrustworthy, and the people who lose most are the ones with no alternative. Whoever solves this is building the trust layer for the next decade of voice technology.

Judging criteria

- ❖ Robustness: how well does the system perform on unseen speakers, engines, and call conditions?
- ❖ Originality: does the solution use signals beyond an off-the-shelf audio classifier?
- ❖ Technical depth: how well-executed is the solution, and does the team understand why it works?
- ❖ Feasibility: could a bank deploy this on real phone audio?
- ❖ Latency: how fast can it determine an outcome with reasonable confidence?